

Flood Resilient Housing Solutions Homes and Communities

Lumbini Province, Nepal
2023-2024

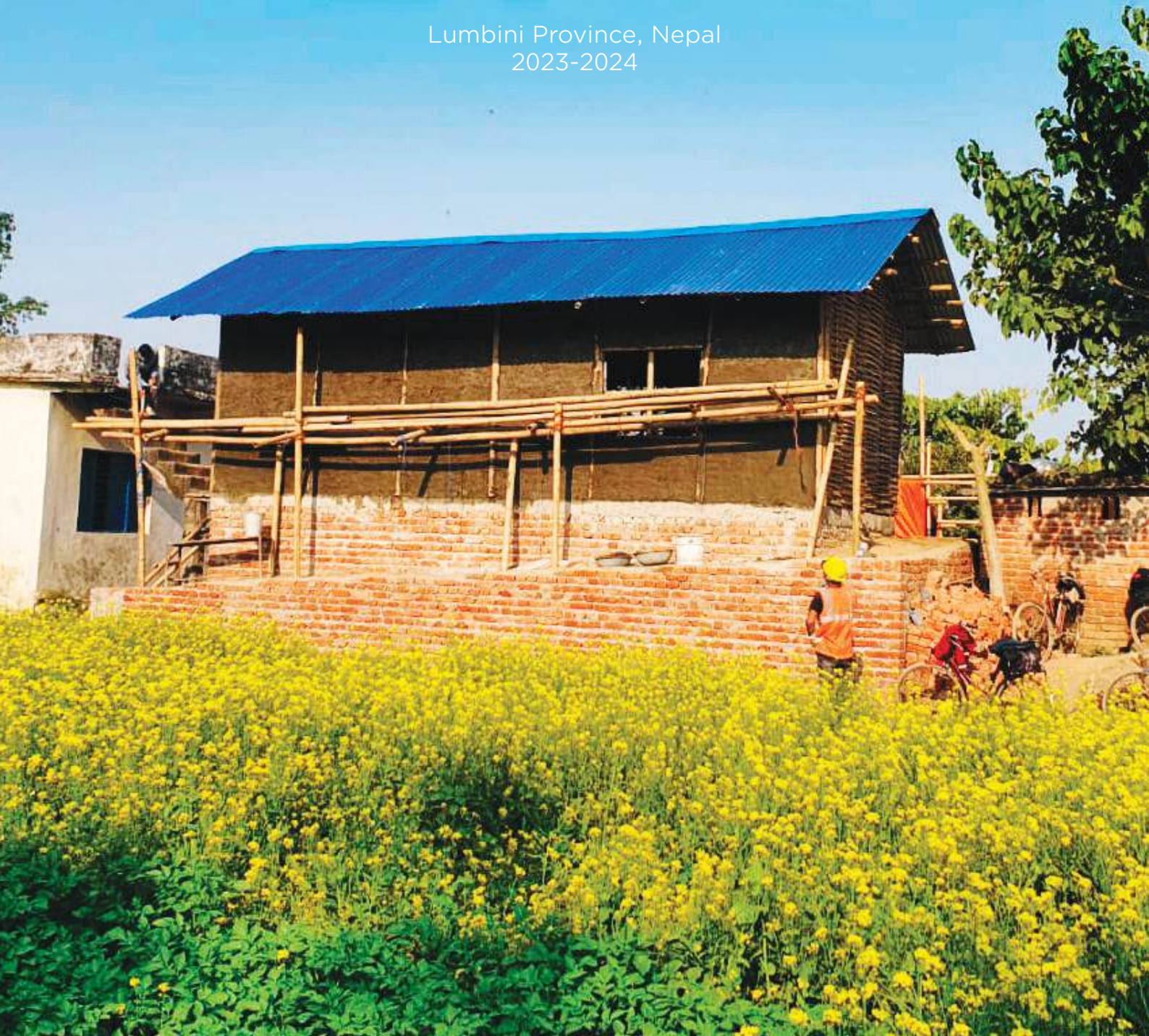


PHOTO STORY



*Building
Resilient
Communities*

Published by:

CRS Nepal

Published date:

2024

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Part 1



Vulnerable Heritage:
Housing typology and its Risk in Banke



In Rapti Sonari Rural Municipality and Duduwa Rural Municipality, traditional housing structures predominantly constructed with wood and bamboo are facing significant challenges due to flood risks. These structures, which embody historical and cultural heritage, often have foundational issues, as many are not elevated above the recommended plinth level. The reliance on wooden posts embedded directly into the soil makes these homes susceptible to rapid deterioration from environmental factors such as moisture, sunlight, and pests. Also, the mud plaster when contact with flood water dissolves rapidly and the house eventually gets tilted causing further damage. Additionally, the roofs of these houses are frequently inadequately secured, increasing their vulnerability to wind damage. Despite the Tharu community's use of locally available resources, traditional construction methods have not evolved to address modern disaster risks effectively. The lack of proper drainage systems exacerbates the situation, particularly impacting economically disadvantaged households and regions prone to flooding. From 2019 to 2022, natural disasters have resulted in the complete destruction of 606 homes, with 14,332 additional structures at risk of flooding.





A traditional home in Rapti Sonari, built with locally sourced wood, mud, and other natural materials, showcases the region's rich vernacular architecture.



A settlement in Duduwa Municipality, nestled right along the riverbank. Its close proximity to the water heightens the community's vulnerability to natural disasters.



The traditional settlement lacks a proper drainage system, leaving it highly vulnerable to flooding



Not only Banke but other districts like Kailali are built on poorly implemented raised ground, which can actually increase their vulnerability to flooding



Traditional houses made of just mud and stones are damaged by the flooding and inundation. Preserving these local structures while introducing innovations that build community resilience is crucial for sustainable development.



Traditionally built houses are cost-effective and made from locally available materials, so any new innovations must adhere to these principles.



Mr. Ram Kumal's traditionally designed family home was damaged by flooding, prompting him to join a mason training program provided by CRS Nepal and Caritas Nepal. His home in the background reveals multiple vulnerabilities, including weakened wood joints and an unstable staircase.



Lack of proper joints and connection in the traditional houses increased the vulnerability to Earthquake and flood.



Mr. Adeel Javaid, Shelter Technical Advisor from CRS conducting assessment and shelter improvement analysis. Assessment shows that local stone masonry construction needs to be improved to withstand floods as well as earthquake which can be done through incorporating lessons from Gorkha Earthquake in mason training.

Part 2



**Building Resilience: Innovating Traditional Housing
with Lime-Stabilized Soil**



Innovation to the already existing traditional methods and vernacular architecture is crucial for municipalities like Rapti Sonari and Duduwa Municipality to effectively combat flooding and other disasters. Recognizing this need, CRS Nepal and local implementing partner Caritas Nepal is collaborating with Rapti Sonari Rural Municipality and Duduwa Rural Municipality to implement Flood Resilient Housing Demonstration Building using lime stabilized soil technology. This innovative approach enables the construction of homes using low-cost and locally available resources such as lime, soil, bamboo, wood, and stone, with the support of skilled local masons and laborers. The technique involves mixing sand, soil, lime, and straw to create a soil-based plasters and floor screeds that is as strong as any other plasters and flooring made with cement and sand. This technology also ensures that houses do not retain moisture during the rainy season, making it a sustainable and economical choice for construction. As part of this initiative, two demo community buildings have been constructed one in Rapti Sonari and one in Duduwa Rural Municipality, showcasing the use of local materials and providing training for local masons.





Ms. Bee Rowan, an expert in lime-stabilized soil, led a training on the cycle of quicklime with students from Bheri Technical School, Nepalgunj, Banke



Students and masons from Banke learn from Ms. Bee Rowan about selecting different types of sand to create quicklime-stabilized soil. Sand improves load-bearing capacity and compressive strength which also reduces cracks in lime stabilized soil plaster.



Trial mix samples of various ratios of quicklime and soil, prepared and displayed for observation



Curing process of lime stabilized trial mix samples to strengthen them, which typically takes 28 days. Curing is done by spraying light mist of water just enough to keep the samples moist and covered with plastic or using jute sacks or clothes.



Labeling and coding various trial mix samples for testing. Each code contains mix design ratio of quick lime : soil : sand : rice husk ash : straw.



Compressive strength test of lime stabilized soil samples to assess their load-bearing capacity.



Wet soaking of lime stabilized soil samples to simulate flooding conditions, showing that the samples remain intact and sturdy even after prolonged contact with water.



A comparative display of lime trial mix samples versus normal non-stabilized mixes, demonstrating how lime stabilized soil can withstand water, while unstabilized normal soil dissolve and erode.



On-the-job training for trainee masons, equipping them with the skills to incorporate innovative techniques in building houses within their community.



Chemical treatment of bamboo with boric acid and borax solution makes bamboo more resistant to insects and fungi by removing starch and killing pests in flood-prone areas.



Dry mixing of straw, rice husk ash, and clayey field tested soil for house plastering



Community-level demonstration of the flood simulation on wall panels.



Flood simulation of wall panels in workshop to assess water resistance and durability. It is done to evaluate if the soil washes away during flooding. Lime stabilized soil wall panels were inundated in water for 14 days before using it in demo-community house. Soil plasters remained intact even after the inundation.



Fire test using a blowtorch on a lime-stabilized soil plastered wall. The walls demonstrate not only flood resistance but also fire resistance.



On-the-job training for trainee masons, featuring a demonstration of setting out for a demo community house in Bairiya Village, Ward No. 01, Rapti Sonari Rural Municipality.



On-the-job mason trainees working to on foundation layout of the demo community house at Rapti Sonari Rural Municipality.



Trainee mason, Mr. Prem Budha digging trenches in Bairiya village for the construction of the demo community building.



Trenches being dug for the foundation of the flood resilient demonstration community building. The size of the building is 24 feet X 12 feet with 2 rooms.



Laying down a raised plinth using locally available- construction materials. It demonstrates how to lay the stones so that there will not be delaminating during earthquake by use of through-stone in the intermediate intervals eg:- 4 feet horizontal and 2 feet vertical.



Raised plinth constructed with concrete Damp Proof Course (DPC) integration. U-strap anchor the timber post with concrete DPC properly.



U-strap inserted into the Damp Proof Course (DPC) / plinth band of the foundation.



Construction of the timber frame for the demo community building. Use of metal straps, horizontal bands, diagonal bracings and vertical posts to withstand earthquake forces.



Construction of the timber frame for the demo community building.



Metal strap being fastened to join timber, steel, and cement elements, crucial for reinforcing the structure.



Trainee masons working on lath construction during their on-the-job training



The demo community building after installing laths



Training on wall panels using the prepared lime stabilized soil mix.



The claw action on the wooden panels after plastering with the dry mix allows the lime-stabilized soil to penetrate the cracks. This strengthens the walls and enhances the house's flood resistance.



Trainee masons engaged in on-the-job training, assembling wattle construction using bamboo mats.



Orientation to masons on better steel reinforcement detailing as recommended by Nepal National Building Code (NBC)



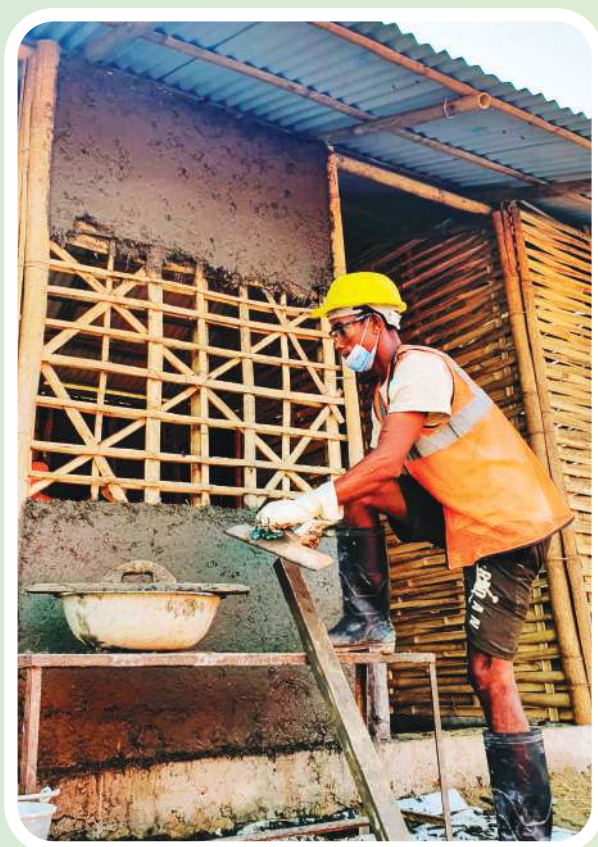
Bamboo wattle in traditional houses utilizes local techniques and materials effectively, but it often lacks durability. Stiffer, stronger wattle is crucial to reducing the vulnerability of these houses.



Interiors of the walls plastered with lime-stabilized soil. The plaster's claw action helps prevent floodwater from entering the home while also strengthening the structure



Plastering lime-stabilized soil in a house to test its efficacy.



Trainee masons participating in on-the-job training to learn plastering techniques using lime-stabilized soil



The flooring Also, the mud plaster when contact with flood water dissolves rapidly and the house eventually gets tilted causing further damage, which helps prevent erosion during floods



Interiors of Rapti Sonari demo community building which used lime stabilized soil in plastering and floors



Curing and shading of the lime-stabilized house after plastering to strengthen the structure.



Exterior view of the Rapti Sonari, Bairiya demo community building following plastering.



Construction of the demo community building in Duduwa. Since the community is unfamiliar with using stone, local masons were trained to work with bricks and other locally available resources.



Raised plinth of the Duduwa demo community building. Considering the high flood levels in the area, the construction elevated the plinth by over 5 feet, highlighting the importance of using local knowledge and tailoring construction to meet the community's needs



Bamboo being used in the Duduwa demo community center instead of timber, as it is more readily available in the area."



Demo-community house at Duduwa Rural Municipality whose plinth height is raised using historical flood data. Flood risk marker on the right side of picture installed by CRS and Caritas shows flood level of 2074 B.S. and 2079 B.S.

Part 3



Community Engagement: Harnessing Local Knowledge and Resources for Disaster-Resilient Communities



CRS Nepal and Caritas Nepal emphasizes the use of local resources and techniques to enhance disaster resilience, working closely with local governments and the National Disaster Risk Reduction and Management Authority. Their projects are rooted in strong community engagement, fostering both direct and indirect relationships with local people and authorities. By incorporating indigenous knowledge and adopting a bottom-up approach, CRS ensures that solutions are inclusive and tailored to community needs. The lime-stabilized technology exemplifies this strategy, using locally sourced materials such as soil, bamboo, straw, sand, and lime to construct disaster-resistant homes. This approach not only reduces costs but also leverages traditional building practices, making it both accessible and sustainable. Community engagement remains at the core of their work.





Community orientation led by the CRS team and Caritas in Bairiya, Rapti Sonari.



Ward Chair Yam Bahadur Khatri participating in the orientation on the demo community house.



Presentation on the demo community house to Narendra Kumar Chaudhary, Chairperson of Duduwa Rural Municipality by CRS.



Students of Bheri Technical School, Nepalgunj receiving a short training on lime-stabilized soil.



On-the-job training for mason workers, teaching them various aspects of constructing, flood-resilient homes.



Team of Bheri Technical school students, CRS team, and trainee masons.



Handover of the demo community building to Mr. Tapta Bahadur Paudel,
Chairperson of Rapti Sonari Rural Municipality.



Handover of the demo community building to Mr. Narendra Kumar Chaudhary,
Chairperson of Duduwa Rural Municipality.

Theory session for trainee masons at Rapti Sonari Rural Municipality by Er. Bibek Nepal, DUDBC Banke

Prototype 3D model of demo-house at Rapti Sonari Rural Municipality featuring raised plinth with earthquake-resilient features



Discussion with community members at Duduwa Rural Municipality by Caritas Nepal regarding the demo community building

Part 4



Empowering Local Builders: Training Masons and Engineers in Lime-Stabilized Soil Construction for Flood-Resilient Homes



In Rapti Sonari Rural Municipality and Gulariya Municipality, twenty-four mason workers recently completed a comprehensive seven-day training on lime-stabilized soil and its application in home construction. The program combined both theoretical instruction and practical experience, building on the knowledge gained during the construction of demo community buildings in Rapti Sonari and Duduwa Rural Municipalities. The masons first acquired a solid understanding of the principles and benefits of using quick lime in construction. They then applied these concepts through hands-on practice, plastering walls of local households using lime-stabilized soil. This dual approach not only equipped the masons with essential skills but also ensured the effective implementation of this innovative technology in building flood-resilient homes for the local community, improving both flood resistance and overall construction quality. In addition to the masons' training, CRS also conducted a three-day training for ten local government engineers from Banke, Bardiya, and Kailali districts, focusing on the use of quicklime in the construction of flood-resilient homes.





Discussions with trainee masons during the practical class



Trainee masons learning how to set up



Trainee masons learning trench excavation and foundation laying techniques



Trainee masons working on constructing the Damp Proof Course (DPC)



Trainee masons visiting the demo community building in Bairiya, Rapti Sonari, to observe lime-stabilized construction techniques



Trainee masons mixing quick lime, straw, rice husk ash, and clay-rich tested soil to prepare a dry mix



Trainee masons learning to plaster wall panels with lime-stabilized soil



Trainee masons using water to cure lime-stabilized plaster, strengthening it in the process



Trainee masons learning about resilient housing construction techniques from Associate Professor Kshitij C. Shrestha of Pulchowk Campus, Institute of Engineering.



Peer-to-peer learning among the trainee masons about bar bending



Discussion with trainee masons

Drying rice husk ash
to be used in lime
stabilization



Bamboo being treated with a
boric and borax acid solution
to make it water-resistant
which enhances insect and
fungal resistance by killing
insects and removing starch,
which attracts pests.

Home plastered by trainee
masons of Rapti Sonari
Rural Municipality using lime
stabilized soil technology as
follow- up activity of training





Trainee masons visiting the demo community center in Bairiya, Rapti Sonari, to observe various flood resilient housing solutions



Engineers from various local government offices observing a demonstration by the CRS team

Engineers observing dry mixing techniques for creating lime-stabilized soil





Group work during Engineer's training in Nepalgunj, Banke on flood resilient housing solutions



Local government engineers visiting demo community house to observe the various flood resilient housing solutions



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